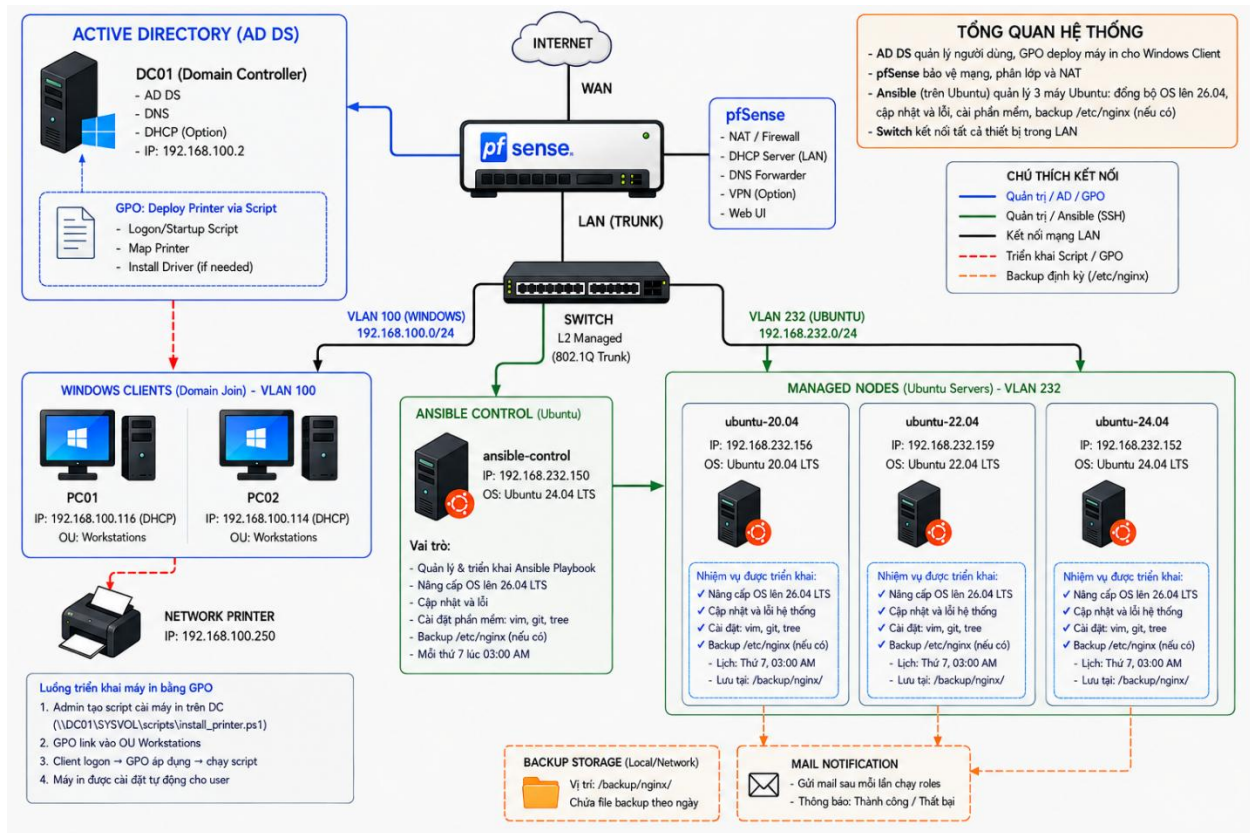


# MỤC LỤC

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# SƠ ĐỒ TỔNG QUAN

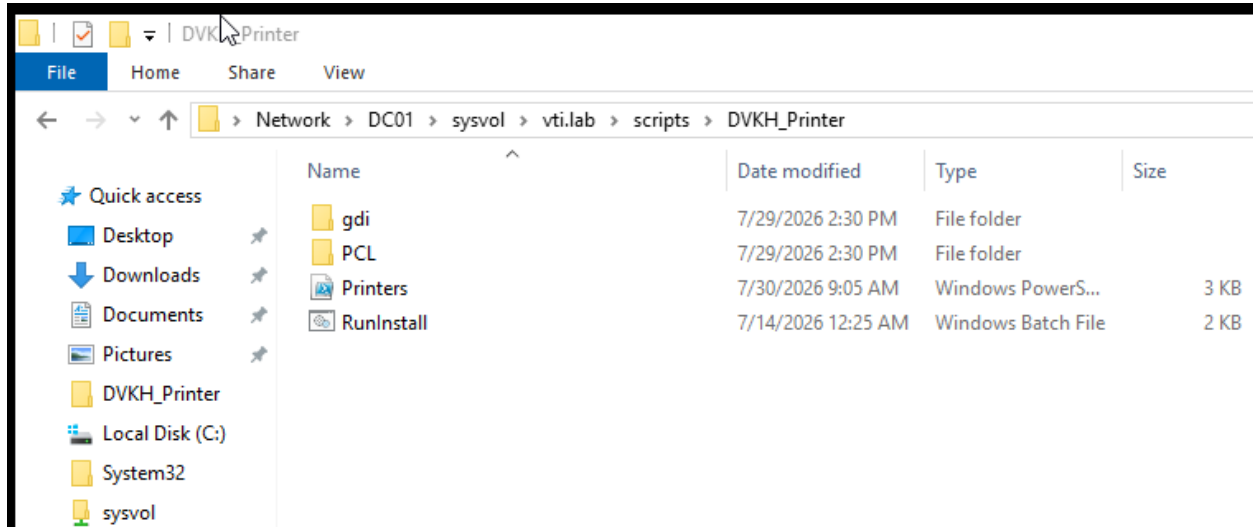


# Quy trình Deploy Printer qua GPO

cấu trúc trên Domain Controller

Tạo thư mục tập trung trên fileserver:

\\vti.lab\SYSVOL\vti.lab\scripts\DVKH\_Printer\



Trong đó:

- Printers.ps1: script chính để cài printer.
- RunInstall.bat: file được GPO gọi khi máy client khởi động.
- gdi\, PCL\ : chứa driver tương ứng với từng dòng máy in.

Printers.ps1:

- Add Driver Printer.
- Add Printer Port.
- Add Printer

# Tạo GPO

Trên Domain Controller:

Group Policy Management



Group Policy Objects

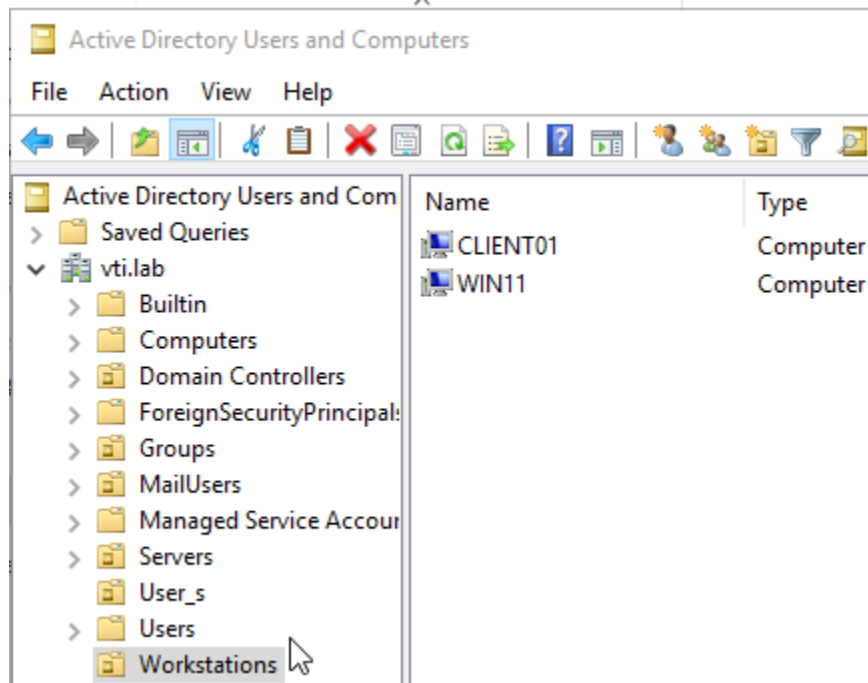


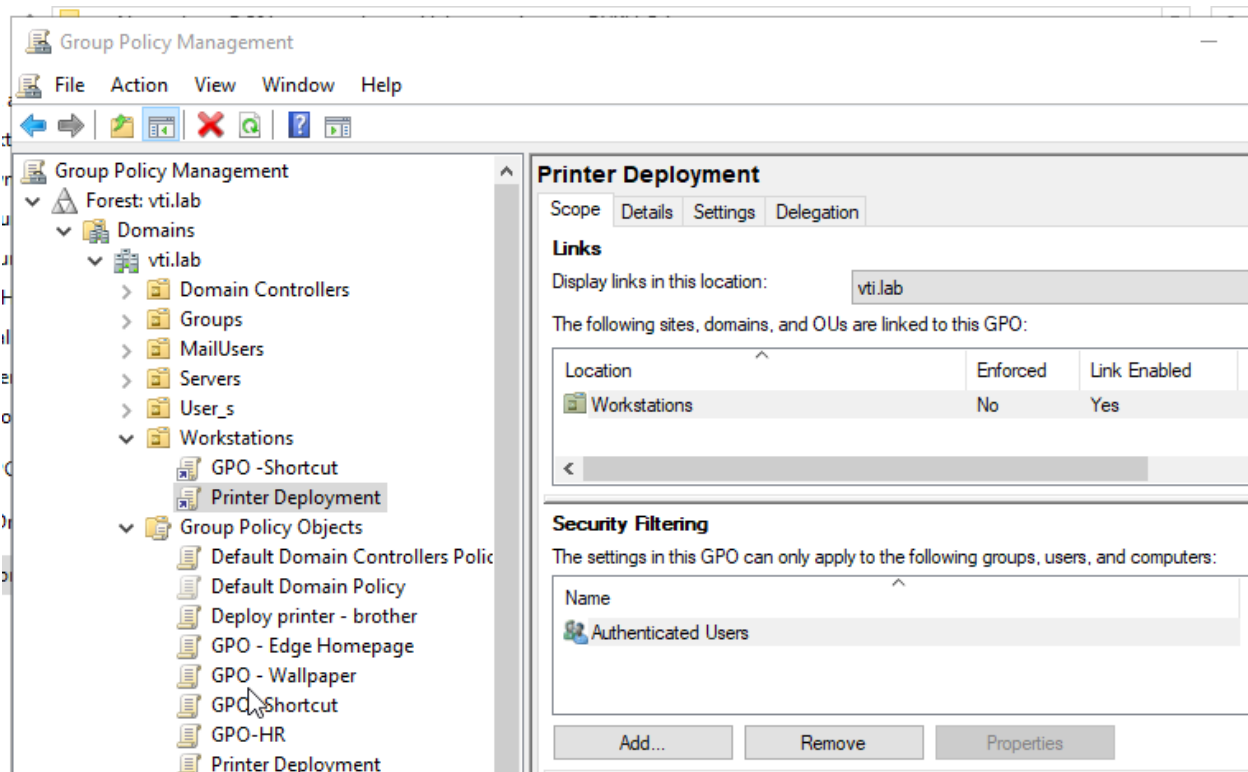
New GPO



Printer Deployment

Link GPO vào OU chứa các máy client cần deploy





## Cấu hình Startup Script

Vào:

Computer Configuration



Policies



Windows Settings

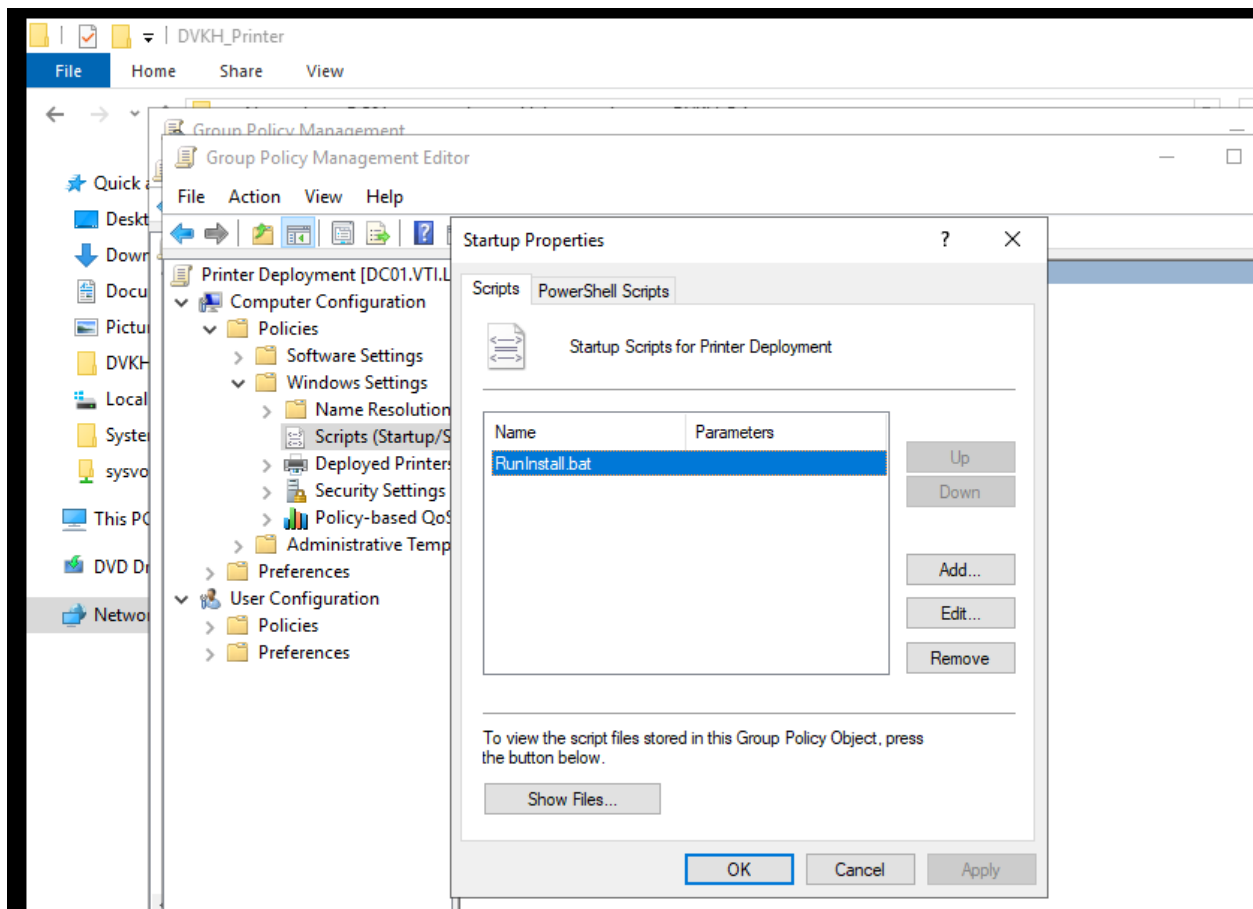


Scripts (Startup/Shutdown)



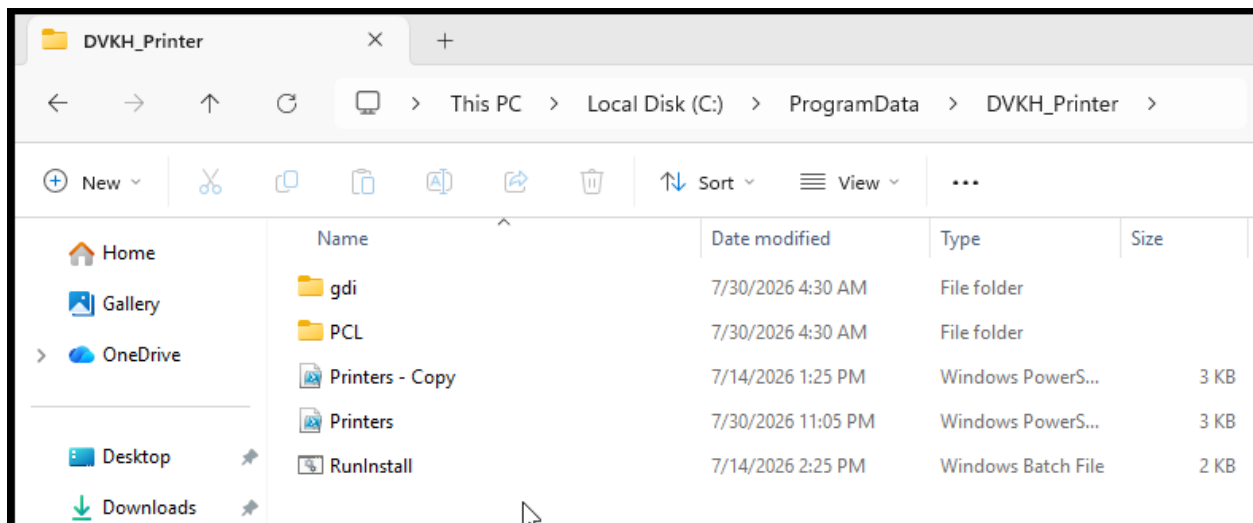
Startup

Thêm file RunInstall.bat



## Copy script xuống máy Client

Do chạy trực tiếp PowerShell từ UNC có thể xuất hiện **Security Warning** nên cần copy script về máy client trước.



Luồng copy file cấu hình từ file server xuống máy client:

SYSVOL

|

| robocopy



C:\ProgramData\DVKH\_Printer

|

|— Printers.ps1

|— gdi\

|— PCL\

Rồi mới chạy file!

## Test GPO

gpupdate /force

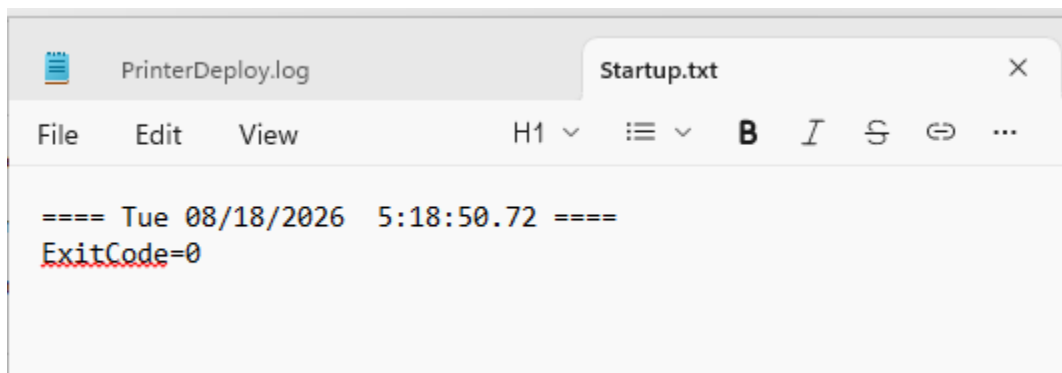
gpresult /r

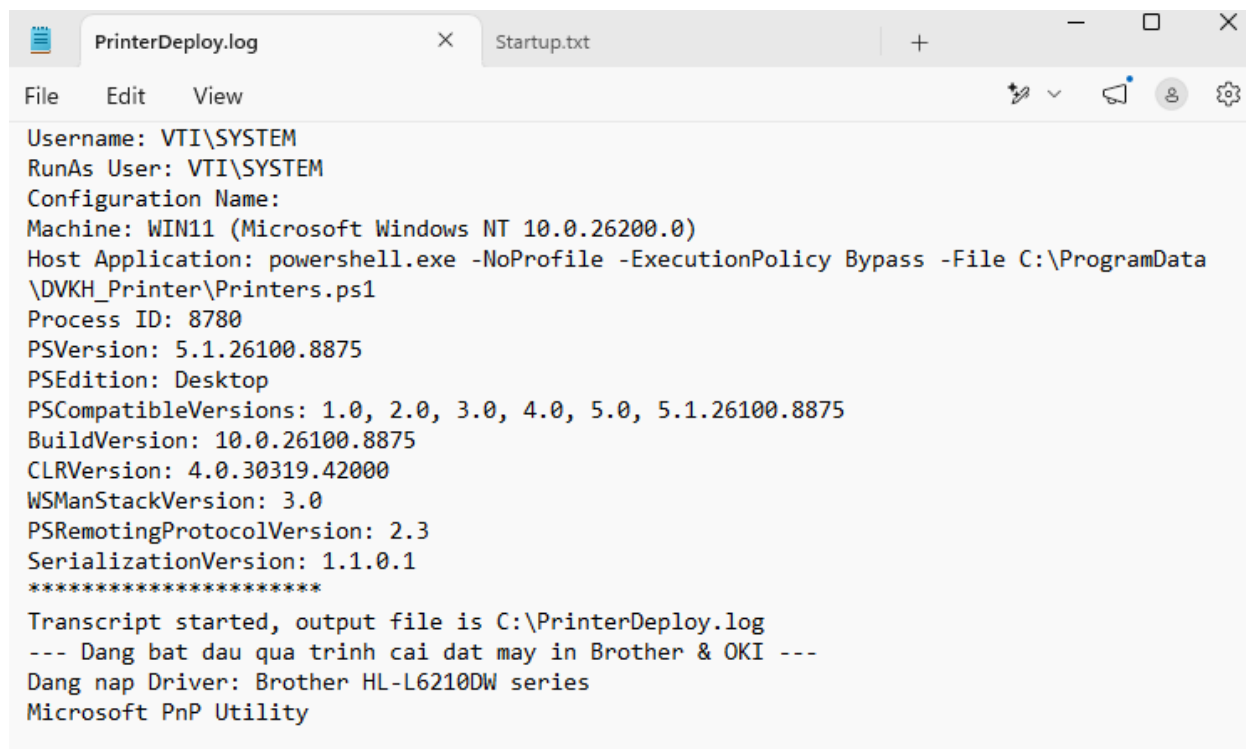
```
C:\Users\hr>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.
```

Test Startup Script có chạy trên máy client không?

Tạo 1 file log để check xem file scrip có chạy trên máy client không?



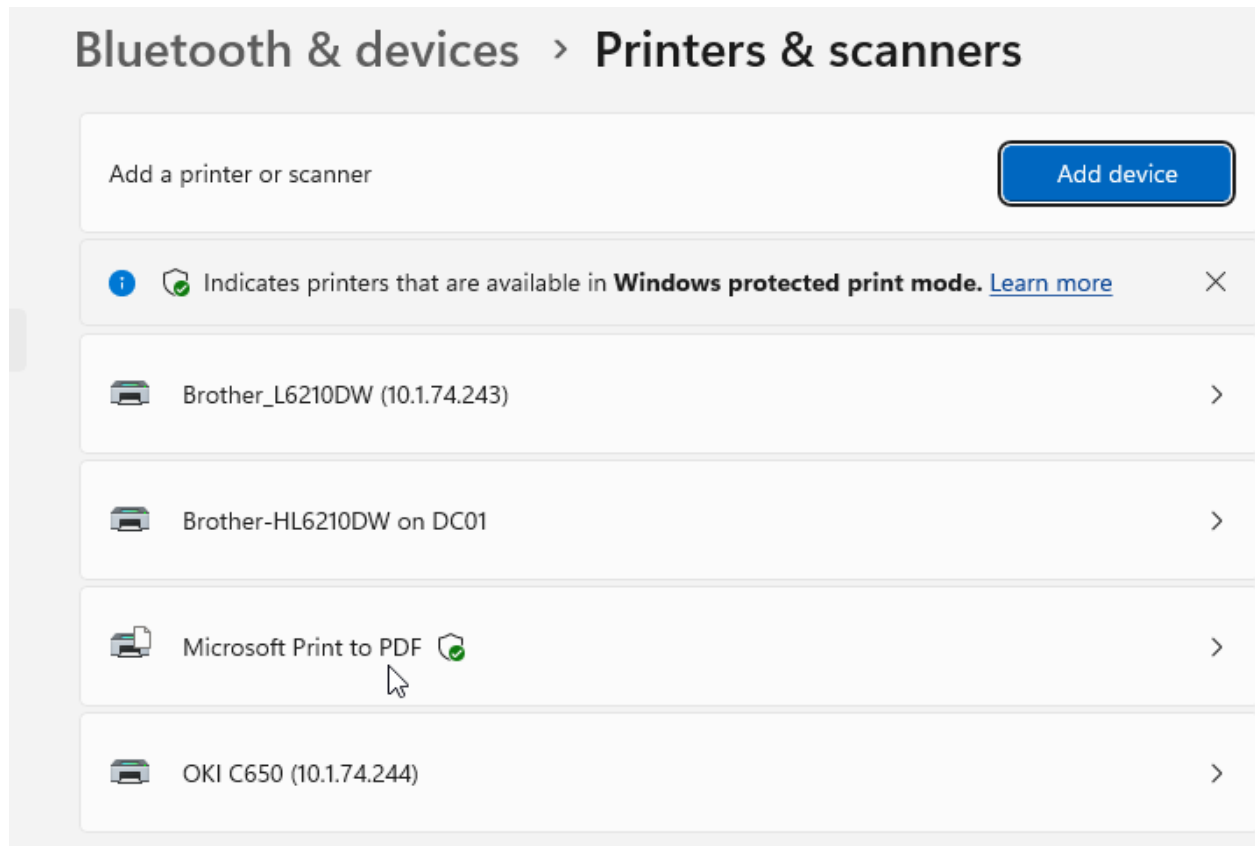


PrinterDeploy.log Startup.txt

File Edit View

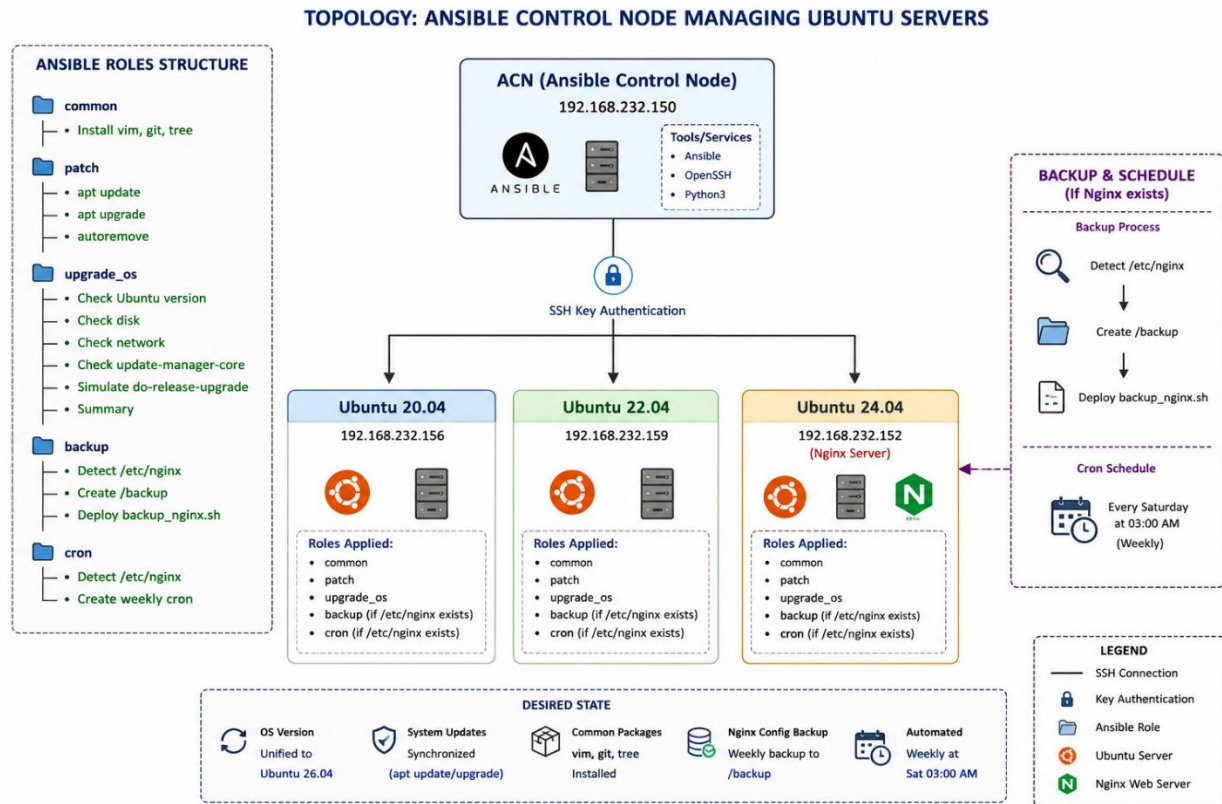
Username: VTI\SYSTEM  
RunAs User: VTI\SYSTEM  
Configuration Name:  
Machine: WIN11 (Microsoft Windows NT 10.0.26200.0)  
Host Application: powershell.exe -NoProfile -ExecutionPolicy Bypass -File C:\ProgramData  
\DVKH\_Printer\Printers.ps1  
Process ID: 8780  
PSVersion: 5.1.26100.8875  
PSEdition: Desktop  
PSCompatibleVersions: 1.0, 2.0, 3.0, 4.0, 5.0, 5.1.26100.8875  
BuildVersion: 10.0.26100.8875  
CLRVersion: 4.0.30319.42000  
WSManStackVersion: 3.0  
PSRemotingProtocolVersion: 2.3  
SerializationVersion: 1.1.0.1  
\*\*\*\*\*  
Transcript started, output file is C:\PrinterDeploy.log  
--- Dang bat dau qua trinh cai dat may in Brother & OKI ---  
Dang nap Driver: Brother HL-L6210DW series  
Microsoft PnP Utility

## Kiểm tra trên client đã cài máy in



Script đã được gọi, máy in đã cài thành công trên client

# Sơ đồ hệ thống Ansible



## Check kết nối ssh tới các máy

```
cluster1@acn:~/ansible-nginx-lab$ ansible all -i inventory.ini -b -K -m command -a "whoami"
BECOME password:
cluster4 | CHANGED | rc=0 >>
root
cluster3 | CHANGED | rc=0 >>
root
cluster5 | CHANGED | rc=0 >>
root
```

```
cluster1@acn:~/ansible-nginx-lab$ sudo tree -L 2
```

```
.
├── backup.yml
├── common.yml
├── cron.yml
├── files
│   ├── custom.conf
│   └── nginx.conf
├── group_vars
│   └── webservers.yml
├── host_vars
│   ├── cluster2.yml
│   └── cluster3.yml
├── install-nginx-lab1.yml
├── install-nginx.yml
├── inventory.ini
├── patch.yml
├── README.md
├── roles
│   ├── backup
│   ├── common
│   ├── cron
│   ├── nginx
│   ├── patch
│   └── upgrade_os
├── site.yml
└── templates
    └── index.html.j2
```

```
12 directories, 15 files
```

## Tạo các role mới bằng Ansible Galaxy

```
cluster1@acn:~/ansible-nginx-lab$ ansible-galaxy init roles/patch
- Role roles/patch was created successfully
cluster1@acn:~/ansible-nginx-lab$ ansible-galaxy init roles/backup
- Role roles/backup was created successfully
cluster1@acn:~/ansible-nginx-lab$ ansible-galaxy init roles/cron
- Role roles/cron was created successfully
cluster1@acn:~/ansible-nginx-lab$ |
```

## Common và test

### Mục tiêu

Cài vim, git, tree cho tất cả server.

```
cluster1@acn:~/ansible-nginx-lab$ sudo vim roles/common/tasks/main.yml
cluster1@acn:~/ansible-nginx-lab$ sudo nano common.yml
```

```
---
- name: Install common packages
  apt:
    name:
      - vim
      - git
      - tree
    state: present
    update_cache: yes
~
```

```
---
- hosts: all
  become: yes

  roles:
    - common
```

```

cluster1@acn:~/ansible-nginx-lab$ ansible-playbook -i inventory.ini common.yml -K
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [cluster4]
ok: [cluster3]
ok: [cluster5]

TASK [common : Install common packages] *****
changed: [cluster4]
changed: [cluster5]
changed: [cluster3]

PLAY RECAP *****
cluster3      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
cluster4      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
cluster5      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

```

Rule common hoạt động đã cài được tree trên 3 máy

```

cluster1@acn:~/ansible-nginx-lab$ ansible all -i inventory.ini -m shell -a "which tree"
cluster4 | CHANGED | rc=0 >>
/usr/bin/tree
cluster5 | CHANGED | rc=0 >>
/usr/bin/tree
cluster3 | CHANGED | rc=0 >>
/usr/bin/tree

```

## Role patch và test

Mục tiêu : update cache, upgrade, xóa unused package

```

cluster1@acn:~/ansible-nginx-lab$ sudo nano common.yml
cluster1@acn:~/ansible-nginx-lab$ sudo vim roles/patch/tasks/main.yml
cluster1@acn:~/ansible-nginx-lab$ sudo vi patch.yml

```

```

---
- name: Update apt cache
  apt:
    update_cache: yes

- name: Upgrade packages
  apt:
    upgrade: dist

- name: Remove unused packages
  apt:
    autoremove: yes

```

```

---
- hosts: all
  become: yes

  roles:
    - patch

```

Chạy playbook

```

cluster1@acn:~/ansible-nginx-lab$ ansible-playbook -i inventory.ini patch.yml -K
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [cluster4]
ok: [cluster5]
ok: [cluster3]

TASK [patch : Update apt cache] *****
changed: [cluster5]
changed: [cluster4]
changed: [cluster3]

TASK [patch : Upgrade packages] *****
changed: [cluster5]
changed: [cluster4]
changed: [cluster3]

TASK [patch : Remove unused packages] *****
ok: [cluster4]
changed: [cluster3]
changed: [cluster5]

PLAY RECAP *****
cluster3      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
cluster4      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
cluster5      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

```

## Role backup và test

Mục tiêu: backup file cấu hình nếu máy có nginx

```

cluster1@acn:~/ansible-nginx-lab$ sudo vim roles/backup/tasks/main.yml
24L, 469B written
cluster1@acn:~/ansible-nginx-lab$ sudo vi backup.yml
[New] 8L, 75B written
cluster1@acn:~/ansible-nginx-lab$ |

```

```

- name: Check nginx directory
  stat:
    path: /etc/nginx
  register: nginx_dir

- name: Show result
  debug:
    msg: "/etc/nginx exists"
  when: nginx_dir.stat.exists

- name: Create backup directory
  file:
    path: /backup
    state: directory
    mode: '0755'
  when: nginx_dir.stat.exists

- name: Backup nginx config
  archive:
    path: /etc/nginx
    dest: "/backup/nginx-{{ ansible_date_time.date }}.tar.gz"
    format: gz
  when: nginx_dir.stat.exists
~

```

```

- hosts: all
  become: yes

  gather_facts: yes

  roles:
    - backup
~

```

Chạy playbook

```

cluster1@acn:~/ansible-nginx-lab$ ansible-playbook -i inventory.ini backup.yml -K
BECOME password:

PLAY [all] *****

```

```

TASK [backup : Show result] *****
ok: [cluster3] => {
  "msg": "/etc/nginx exists"
}
skipping: [cluster4]
skipping: [cluster5]

TASK [backup : Create backup directory] *****
skipping: [cluster4]
skipping: [cluster5]
changed: [cluster3]

TASK [backup : Backup nginx config] *****
skipping: [cluster4]
skipping: [cluster5]
changed: [cluster3]

PLAY RECAP *****
cluster3      : ok=5    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
cluster4      : ok=2    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
cluster5      : ok=2    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0

```

Do chỉ máy cluster 3 có nginx nên test thành công!

Server đã tạo file back up thành công trên máy cluster3

```

cluster3@web2:~$ ls /etc/nginx
conf.d      koi-utf    modules-available  nginx.conf.bak  sites-available  uwsgi_params
fastcgi.conf  koi-win    modules-enabled    proxy_params    sites-enabled    win-utf
fastcgi_params mime.types  nginx.conf         scgi_params     snippets

cluster3@web2:~$ ls /backup
nginx-2026-08-06.tar.gz
cluster3@web2:~$

```

## Role cron

Mục tiêu : cron chạy 03:00 sáng thứ Bảy

```

cluster1@acn:~/ansible-nginx-lab$ sudo vim roles/cron/tasks/main.yml
[sudo] password for cluster1:
cluster1@acn:~/ansible-nginx-lab$ sudo vim cron.yml

```

```

- name: Check nginx directory
  stat:
    path: /etc/nginx
  register: nginx_dir

- name: Create cron for nginx backup
  cron:
    name: Weekly nginx backup
    minute: "0"
    hour: "3"
    weekday: "6"
    job: "tar czf /backup/nginx-$(date +\\%F).tar.gz /etc/nginx"
    when: nginx_dir.stat.exists

```

```

---
- hosts: all
  become: yes

  roles:
    - cron

```

```

cluster1@acn:~/ansible-nginx-lab$ ansible-playbook -i inventory.ini cron.yml -K
BECOME password:

```

Chạy playbook

```

TASK [cron : Create cron for nginx backup] *****
skipping: [cluster4]
skipping: [cluster5]
changed: [cluster3]

PLAY RECAP *****
cluster3 : ok=3  changed=1  unreachable=0  failed=0  skipped=0  rescued=0  ignor
ed=0
cluster4 : ok=2  changed=0  unreachable=0  failed=0  skipped=1  rescued=0  ignor
ed=0
cluster5 : ok=2  changed=0  unreachable=0  failed=0  skipped=1  rescued=0  ignor
ed=0

```

Kiểm tra thư mục back up

```

cluster1@acn:~/ansible-nginx-lab$ ansible all -i inventory.ini -b -K -m shell -a "crontab -l"
BECOME password:
cluster4 | FAILED | rc=1 >>
no crontab for rootnon-zero return code
cluster5 | FAILED | rc=1 >>
no crontab for rootnon-zero return code
cluster3 | CHANGED | rc=0 >>
#Ansible: Weekly nginx backup
0 3 * * 6 tar czf /backup/nginx-$(date +%F).tar.gz /etc/nginx
cluster1@acn:~/ansible-nginx-lab$

```

Nếu chạy lại thì vẫn hiện thi ok thay vì change

```

TASK [Gathering Facts] *****
ok: [cluster4]
ok: [cluster5]
ok: [cluster3]

TASK [cron : Check nginx directory] *****
ok: [cluster4]
ok: [cluster5]
ok: [cluster3]

TASK [cron : Create weekly backup cron] *****
skipping: [cluster4]
skipping: [cluster5]
ok: [cluster3]

PLAY RECAP *****
cluster3      : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignor
ed=0
cluster4      : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignor
ed=0
cluster5      : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignor
ed=0

```

Kiểm tra role back up

```

cluster1@acn:~/ansible-nginx-lab$ ansible cluster3 -i inventory.ini -b -K \
-m shell -a "ls -l /usr/local/bin/backup_nginx.sh"
BECOME password:
cluster3 | CHANGED | rc=0 >>
-rwxr-xr-x 1 root root 131 Aug  6 10:01 /usr/local/bin/backup_nginx.sh

```

Kiểm tra cron

```

cluster1@acn:~/ansible-nginx-lab$ ansible cluster3 -i inventory.ini -b -K \
-m shell -a "crontab -u root -l"
BECOME password:
cluster3 | CHANGED | rc=0 >>
#Ansible: Weekly nginx configuration backup
0 3 * * 6 /usr/local/bin/backup_nginx.sh

```

## Roles update OS

Check version OS

```

cluster1@acn:~/ansible-nginx-lab$ ansible all -i inventory.ini -m setup -a "filter=ansible_distribution_version"
cluster4 | SUCCESS => {
  "ansible_facts": {
    "ansible_distribution_version": "20.04",
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false
}
cluster5 | SUCCESS => {
  "ansible_facts": {
    "ansible_distribution_version": "22.04",
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false
}
cluster3 | SUCCESS => {
  "ansible_facts": {
    "ansible_distribution_version": "24.04",
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false
}

```

```
cluster1@acn:~/ansible-nginx-lab$ tree roles/upgrade_os
roles/upgrade_os
├── defaults
│   └── main.yml
├── files
├── handlers
│   └── main.yml
├── meta
│   └── main.yml
├── README.md
├── tasks
│   └── main.yml
├── templates
├── tests
│   ├── inventory
│   └── test.yml
└── vars
    └── main.yml
```

Tạo role updae OS

```
cluster1@acn:~/ansible-nginx-lab$ sudo vim roles/upgrade_os/defaults/main.yml
```

```
---
target_ubuntu_version: "26.04"
simulate_upgrade: true
```

```
cluster1@acn:~/ansible-nginx-lab$ sudo vim roles/upgrade_os/tasks/main.yml
```

```

- name: Show current Ubuntu version
  debug:
    msg:
      - "Hostname : {{ inventory_hostname }}"
      - "Current OS : Ubuntu {{ ansible_distribution_version }}"
      - "Target OS : Ubuntu {{ target_ubuntu_version }}"

- name: Check free disk space
  shell: df -h /
  register: disk_usage
  changed_when: false

- name: Show disk usage
  debug:
    var: disk_usage.stdout_lines

- name: Check Internet connectivity
  shell: ping -c 2 archive.ubuntu.com
  register: network_test
  ignore_errors: yes
  changed_when: false

- name: Show network status
  debug:
    msg: >-
    {{
      'Internet connection OK'
      if network_test.rc == 0
      else 'Internet connection FAILED'
    }}

- name: Determine whether upgrade is required
  set_fact:
    upgrade_required: "{{ ansible_distribution_version != target_ubuntu_version }}"

- name: Simulate Release Upgrade
  debug:
    msg:
      - "[SIMULATION] apt update"

```

Cấu hình lại site. yml : thêm tag

```
cluster1@acn:~/ansible-nginx-lab$ sudo vi site.yml
```

```
---
- name: System Management
  hosts: all
  become: yes
  gather_facts: yes

  roles:
    - { role: common, tags: common }
    - { role: patch, tags: patch }
    - { role: upgrade_os, tags: upgrade_os }
    - { role: backup, tags: backup }
    - { role: cron, tags: cron }
```

Chạy riêng roles update OS

```
cluster1@acn:~/ansible-nginx-lab$ ansible-playbook -i inventory.ini site.yml -K --tags upgrade_os
BECOME password:

PLAY [System Management] *****

TASK [Gathering Facts] *****
ok: [cluster4]
ok: [cluster5]
ok: [cluster3]

TASK [upgrade_os : Show current Ubuntu version] *****
ok: [cluster3] => {
  "msg": [
    "Hostname : cluster3",
    "Current OS : Ubuntu 24.04",
    "Target OS : Ubuntu 26.04"
  ]
}
ok: [cluster4] => {
  "msg": [
    "Hostname : cluster4",
    "Current OS : Ubuntu 20.04",
    "Target OS : Ubuntu 26.04"
  ]
}
ok: [cluster5] => {
  "msg": [
    "Hostname : cluster5",
```

```

TASK [upgrade_os : Check free disk space] *****
ok: [cluster4]
ok: [cluster5]
ok: [cluster3]

TASK [upgrade_os : Show disk usage] *****
ok: [cluster4] => {
  "disk_usage.stdout_lines": [
    "Filesystem                Size      Used Avail Use% Mounted on",
    "/dev/mapper/ubuntu--vg-ubuntu--lv 9.8G  5.3G  4.0G  57% /"
  ]
}
ok: [cluster3] => {
  "disk_usage.stdout_lines": [
    "Filesystem                Size      Used Avail Use% Mounted on",
    "/dev/mapper/ubuntu--vg-ubuntu--lv 9.8G  5.0G  4.3G  54% /"
  ]
}
ok: [cluster5] => {
  "disk_usage.stdout_lines": [
    "Filesystem                Size      Used Avail Use% Mounted on",
    "/dev/mapper/ubuntu--vg-ubuntu--lv 9.8G  5.2G  4.2G  56% /"
  ]
}

```

```

TASK [upgrade_os : Check Internet connectivity] *****
ok: [cluster4]
ok: [cluster5]
ok: [cluster3]

TASK [upgrade_os : Show network status] *****
ok: [cluster3] => {
  "msg": "Internet connection OK"
}
ok: [cluster4] => {
  "msg": "Internet connection OK"
}
ok: [cluster5] => {
  "msg": "Internet connection OK"
}

TASK [upgrade_os : Determine whether upgrade is required] ****
ok: [cluster3]
ok: [cluster4]
ok: [cluster5]

```

```
TASK [upgrade_os : Simulate Release Upgrade] *****
ok: [cluster3] => {
    "msg": [
        "[SIMULATION] apt update",
        "[SIMULATION] apt upgrade",
        "[SIMULATION] install update-manager-core",
        "[SIMULATION] do-release-upgrade",
        "[SIMULATION] reboot",
        "[SIMULATION] wait_for_connection"
    ]
}
ok: [cluster4] => {
    "msg": [
        "[SIMULATION] apt update",
        "[SIMULATION] apt upgrade",
        "[SIMULATION] install update-manager-core",
        "[SIMULATION] do-release-upgrade",
        "[SIMULATION] reboot",
        "[SIMULATION] wait_for_connection"
    ]
}
ok: [cluster5] => {
    "msg": [
        "[SIMULATION] apt update",
        "[SIMULATION] apt upgrade",
        "[SIMULATION] install update-manager-core",
        "[SIMULATION] do-release-upgrade",
        "[SIMULATION] reboot",
        "[SIMULATION] wait_for_connection"
    ]
}
```

```

TASK [upgrade_os : Finish] *****
ok: [cluster3] => {
  "msg": "Upgrade simulation completed."
}
ok: [cluster4] => {
  "msg": "Upgrade simulation completed."
}
ok: [cluster5] => {
  "msg": "Upgrade simulation completed."
}

PLAY RECAP *****
cluster3           : ok=9    changed=0    unreachable=0
cluster4           : ok=9    changed=0    unreachable=0
cluster5           : ok=9    changed=0    unreachable=0

```

## Notification

Chạy riêng lẻ từng role

```

ansible-playbook -i inventory.ini site.yml -K \
  --vault-password-file ~/.ansible_vault_pass \
  --tags common \ -e "notification_role=common"

```

**[ANSIBLE] common - SUCCESS** Hộp thư đến x

**hungdoansky@gmail.com**

đến tôi ▼

```

=====
      ANSIBLE UBUNTU LAB REPORT
=====

```

ROLE:  
common

DATE:  
2026-08-20

TIME:  
03:18:29

=====

## HOST RESULTS

=====

Host:

cluster3

IP:

192.168.232.152

OS:

Ubuntu

Version:

24.04

STATUS:

SUCCESS